Name: ________________________, Roll No.: ____________

Total marks:

CSE 350/550: Network Security
Quiz 1: TUE Feb 04, 2025

Instructions:

- There are 4 questions, each 7 marks, for a total of 28 marks. Total time available 30 minutes.
- Provide answers only in the space provided. You may do your rough work on a separate blank sheet.
- You may use a calculator, but nothing else. **Pl. keep your phones/laptops, books etc. away from you.**
- Do not use unfair means, else action will be taken. Do not use material such as books, notes, etc. or a computer, smart phone, etc. And do not share information with other students.

Marks, Q1:

1. Consider a crypto-system similar to Caesar-cipher, but where c=p*k mod 26. Here the set of symbols is {0, 1, 2, ..., 24, 25}, p is plaintext, c is ciphertext, k is key. For example, if key=3, p=15, then c=19. Is there a constraint on the value of key, k? If so, what condition must k satisfy to qualify as a valid key. And give two examples of k that are NOT allowed, and another 2 examples of k that are allowed.

4

3 marks

Condition that k must satisfy to qualify as a valid key:
k must be such that 26 & k are co-prime. That is k should not have 2 or 13 as a factor.

2 marks each

2 examples of k that are NOT allowed: i ii ~~2, 4, 6~~ any two of 2, 4, 6, 8, 10, 12 14, 16, 18, 20, 22, 24 and 13

2 examples of k that are allowed: i ii Any two of 3, 5, 7, 9, 11, 15, 17 19, 21, 23.

Marks, Q2:

2. Consider the poly-alphabetic substitution cipher discussed in class, where the key is 'deceptive'. If the **ciphertext** is
z l j e k x j z l q h k w r h d z v h h
then what is the plaintext? Show your working in the table below.

key	d	e	c	e	p	t	i	v	e	d	e	c	e	p	t	i	v	e	w	e
numeric key	3	4	2	4	15	19	8	21	4	3	4	2	4	15	19	8	21	4	3	4
plaintext	w	e	h	a	v	e	b	e	e	n	d	i	s	c	o	v	e	r	e	d
ciphertext	z	i	j	e	k	x	j	z	i	q	h	k	w	r	h	d	z	v	h	h

Plaintext: ______________________________

4 marks for correct answer
3 marks for completing the table

3. The initial permutation (before round 1) in DES is described by an 8x8 matrix. This is given below. The final permutation (after round 16) reverses the permutation carried out in the initial permutation. This is also described by an 8x8 matrix, but only a small portion of it is shown below. It also shows that the 1-st output bit is the 40-th input bit to the final permutation. In what way are the 2-nd, 3-rd, ..., 8-th output bits related to the input bits of the final permutation? Just fill in the blank portions of the table.

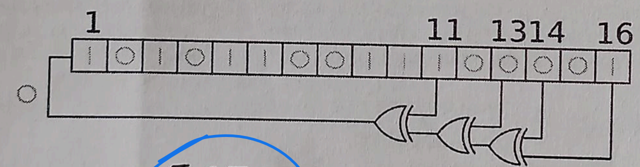
Marks, Q3:

1 mark each

Initial Permutation	Final Permutation
58 50 42 34 26 18 10 02	40 8 48 16 56 24 64 32
60 52 44 36 28 20 12 04	
62 54 46 38 30 22 14 06	
64 56 48 40 32 24 16 08	
57 49 41 33 25 17 09 01	
59 51 43 35 27 19 11 03	
61 53 45 37 29 21 13 05	
63 55 47 39 31 23 15 07	

4. Consider the operation of a pseudo-random number generator based on a 16-bit linear feedback shift register shown below. What are the first THREE 16-bit random numbers generated following 1010 1100 1110 0001 or 'ACE1' in Hexadecimal digits? Bit 1 is most significant, while bit 16 is least significant. Write your answer as Hexadecimal digits.

Marks, Q4:



Random no. i 5670

Random no. ii AB38

Random no. iii 559C

or

0 101 0110 0111 0000

1010 1011 0011 1000

0101 0101 1001 1100

2 marks each

+ 1 mark if all are correct